

Pressure Movement Sensation with Rack and Pinion Based Wearable Interface

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Abstract

We propose a novel haptic device, enabling a pressure movement sensation on the user's hand for immersive VR hand interaction. To render pressure movement on the hand, we examine the perceptual characteristics of pressure-based phantom sensation. Unlike vibration-based phantom sensation, pressure-based moving phantom sensation has received limited attention in prior research. In this work, we present a novel haptic device and rendering method specifically designed to induce pressure-based moving phantom sensation. This unique form of feedback allows users to experience more realistic interactions in virtual environments, such as handshakes.

CCS Concepts

• **Human-centered computing** → **Haptic devices**; **Human computer interaction (HCI)**; *User studies*; *Virtual reality*.

Keywords

pressure feedback, tactile illusion, wearables, haptics

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1 INTRODUCTION

In virtual environments, users primarily interact through their hands. As such, developing haptic devices that can deliver realistic tactile sensations to the hand is a critical task in haptic research. Providing appropriate haptic feedback on the hand not only enhances immersion but also strengthens the realism within virtual

experiences. Among hand-based interactions, handshakes evoke a meaningful emotional interaction [4], and several devices [5, 7] have been developed to simulate this handshake. Building upon prior work, we present a novel wearable device capable of conveying dynamic pressure sensations on the ulnar-sided hand.

To convey the sensation of moving pressure, our study leverages the concept of phantom sensation. Unlike vibration-based feedback, delivering pressure feedback typically requires bulky actuators [3]. However, by inducing pressure-based phantom sensations, our approach allows for more efficient device design. Moreover, phantom sensations related to pressure remain significantly underexplored. Prior studies have predominantly focused on vibration-based phantom sensations [1], and even in the context of pressure, most research has been limited to the forearm [2, 6]. In contrast, our work integrates this tactile illusion into the hand, aiming to deliver a more natural movement of pressure through a compact, wearable form factor.

Furthermore, implementing pressure-based moving phantom sensation can be connected to new opportunities in haptic research. The proposed approach not only expands the applicability of existing pressure-based haptic devices to a wider range of scenarios but also offers a foundation for future designs to achieve greater hardware efficiency. In addition, our wearable device allows for a free-hand configuration, enabling seamless integration with other haptic systems. Thus, our haptic interface can enhance the overall sense of immersion in virtual environments.

2 PRESSURE MOVEMENT HAPTIC DEVICE

Delivering pressure feedback through haptic devices typically requires bulky hardware configurations. Motor-based systems allow for precise control but tend to be heavy and overly constrain the user's hand movement. Pneumatic approaches offer a lighter form factor, but they require external air pumps. To address these limitations and render pressure more efficiently, we adopt a tactile illusion technique previously used in haptic rendering. This approach enables the delivery of feedback to locations without physical actuators, thereby overcoming the conventional limitation of spatially constrained feedback. As a result, our system can provide pressure sensations over a broader area using fewer actuators.

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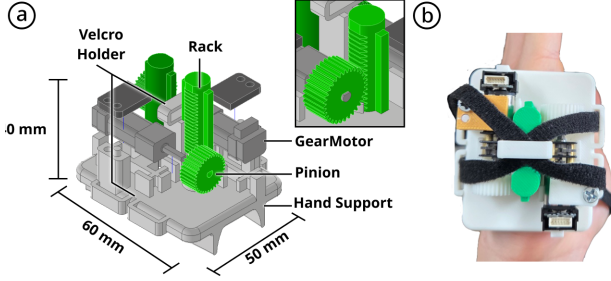


Figure 1: (a) The device uses a rack-and-pinion mechanism to generate pressure feedback. (b) The image shows a user wearing our haptic device.

To ensure precise control of indentation depth and guarantee non-backdrivability during pressure feedback, we adopted a rack-and-pinion mechanism. As the pinion rotates, it drives the linear motion of the rack. This motion is controlled based on motor encoder feedback, allowing accurate modulation of the rack positions, the indentation depth. The device was designed using 3D printing and weighs approximately 50 g. Each actuator applies pressure with a contact diameter of 10 mm. By varying the indentation depth between the two actuators, we were able to generate the sensation of pressure between the two actuators.

Our device is mounted on the ulnar-sided hand using a velcro strap that secures the device. As the motor drives the rack and pinion, the indentation depths are dynamically adjusted to render pressure movement. The user tends to perceive pressure toward the actuator with the greater indentation depth. By continuously modulating the relative depths between the two actuators, the system creates the illusion of pressure movement across the skin.

3 DEMO APPLICATION

Pressure movement occurs either when the user moves their hand or when external objects move across the user's hand. Since such interactions are common in virtual environments, realistically rendering the motion of pressure is essential for enhancing the quality of the VR experience. Our proposed device targets the ulnar-sided hand, making it particularly well-suited for scenarios such as virtual handshakes. We anticipate that this approach can be extended to other parts of the body, enabling the use of pressure-based moving phantom sensation in a wider range of interaction scenarios.

We designed two distinct scenarios. The first is a handshake scenario, where users engage in a virtual handshake with an avatar and perceive the sensation of being grasped through our device. The second is a hammering scenario. As users drive a virtual nail using their hand's ulnar side, the system delivers pressure feedback at the point of contact, with intensity modulated according to the virtual nail's depth. These scenarios illustrate how our device can enhance both immersion and realism.

In this demo, the user wears the proposed haptic device secured with Velcro and a VR HMD. When the user reaches out and performs a handshake gesture toward a rendered avatar in the virtual scene, a corresponding animation is triggered. The user's hand is tracked in real time, and haptic feedback is provided accordingly.

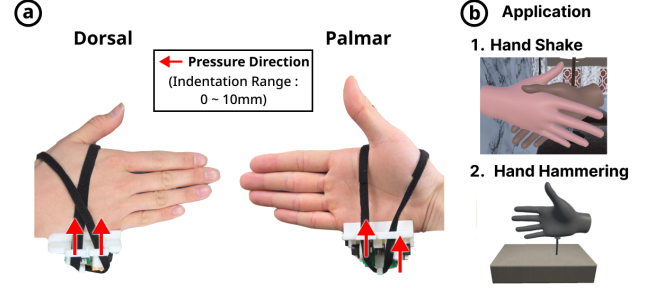


Figure 2: (a) The user wears the device on their right hand and (b) engages in VR interaction scenarios while experiencing the corresponding haptic feedback.

The device renders a pressure movement sensation that aligns with the finger movement of the avatar's hand—from the index to the pinky finger—allowing the user to perceive the sensation of being grasped. In the subsequent scene, the user engages in a hammering task using a virtual metal hand. As the nail contacts the surface and penetrates the table, the user feels pressure feedback that corresponds to the contact point and depth of penetration, experiencing a dynamic change in pressure sensation.

4 CONCLUSION

This demo shows the phantom sensation in the context of pressure and introduces a wearable haptic device designed for use in VR interaction scenarios. By modulating the indentation depths of two actuators, our system simulates dynamic tactile sensations. We designed a haptic device and corresponding VR scenarios that leverage pressure-based moving phantom sensation. Our findings suggest that this approach enables realistic pressure movement, offering new possibilities for immersive haptic experiences.

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